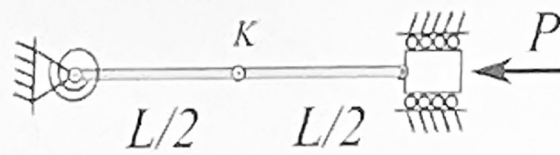
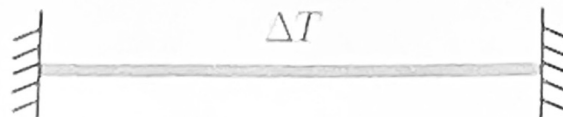


(1.) The structure shown below consists of two rigid members of length $L/2$ connected via pins and one torsion spring. Use K as the spring's stiffness constant. What is the critical buckling load?

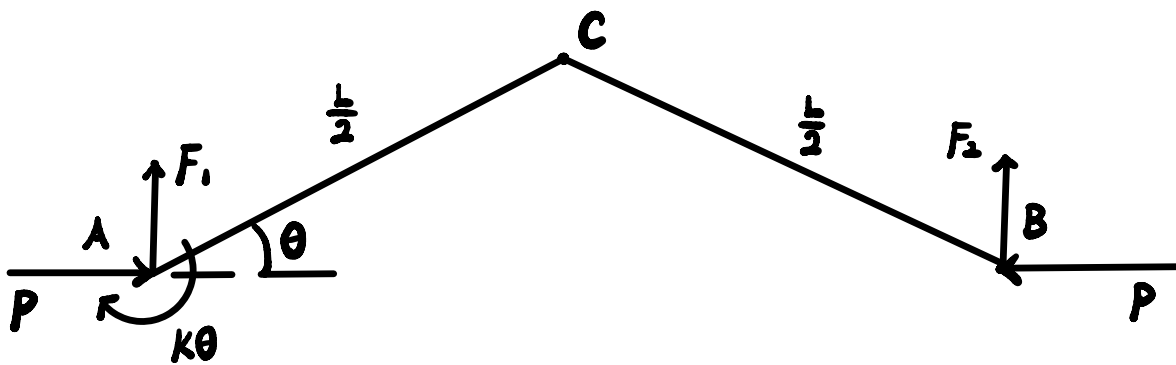


(2.) A fixed-fixed strut of rectangular cross section $2 \text{ cm} \times 10 \text{ cm}$, and 2 m length was made of EN9 carbon steel. Properties of EN9 steel: $E = 200 \text{ GPa}$, $\sigma_y = 350 \text{ MPa}$, thermal expansion coefficient: $\alpha = 11.6 \times 10^{-6} \text{ 1/K}$.

- What is the expected failure mode under compressive loading?
- Calculate ΔT that would trigger Euler buckling failure.



1. At buckling, $P = P_{\text{crit}}$, where θ is small



$$\sum F_v = F_1 + F_2 = 0 \quad \therefore F_1 = -F_2$$

$$\sum M_A = K\theta - F_2 L = 0 \quad \therefore F_2 = \frac{K\theta}{L}$$

$$\therefore F_1 = -\frac{K\theta}{L}$$

$$\text{At } B, \Sigma F_v = F_2 + T_{BC} \sin \theta = F_2 + T_{BC} \theta = 0$$

$$\Sigma F_h = P + T_{BC} \cos \theta = P + T_{BC} = 0$$

$$\therefore T_{BC} \theta = -\frac{K\theta}{L} \quad \therefore T_{BC} = -\frac{K}{L}$$

$$\therefore P_{crit} = -T_{BC} = \frac{K}{L}$$

$$2. \quad r = \sqrt{\frac{I}{A}} = \frac{43}{60} 5.774 \times 10^{-3}$$

$$\frac{0.5L}{r} = \frac{L_{eff} = \frac{1}{2}L \text{ for fixed-fixed}}{r} = \frac{4013}{173.21}$$

$$R_{trans} = \sqrt{\frac{2\pi^2 E}{6\gamma}} = 106.21$$

$$\therefore \frac{L}{r} > R_{trans} \quad \therefore \text{plastic deformed}$$

$$\sigma_{crit} = \frac{\pi^2 EI}{\frac{1}{4} L^2 A} = \frac{\pi^2 E}{(\frac{L}{2r})^2} = \frac{65.8}{41} \text{ MPa}$$

$$\therefore \sigma_{crit} = \frac{F}{A} + E\alpha\Delta T = 41 \text{ MPa} + E\alpha\Delta T$$

$$\text{When } \left(\frac{L}{r}\right)^2 = R_{trans},$$

$$\sigma_{crit}' = \frac{\pi^2 E}{R_{trans}^2} = 175 \text{ MPa}$$

$$\therefore \Delta T = 28.36 \text{ K} = 28.36^\circ \text{C}$$